

EXPERIMENT-21 DATA PREPROCESSING AND ANALYSIS FOR DATASET USING WEKA

The screenshot shows the Weka Explorer window with the 'Visualize' tab active. The 'Current relation' is 'iris' with 150 instances and 5 attributes. The 'Attributes' list on the left shows 'sepalength', 'sepalwidth', 'petalength', 'petalwidth', and 'class'. The 'Selected attribute' table on the right shows the distribution of the 'class' attribute:

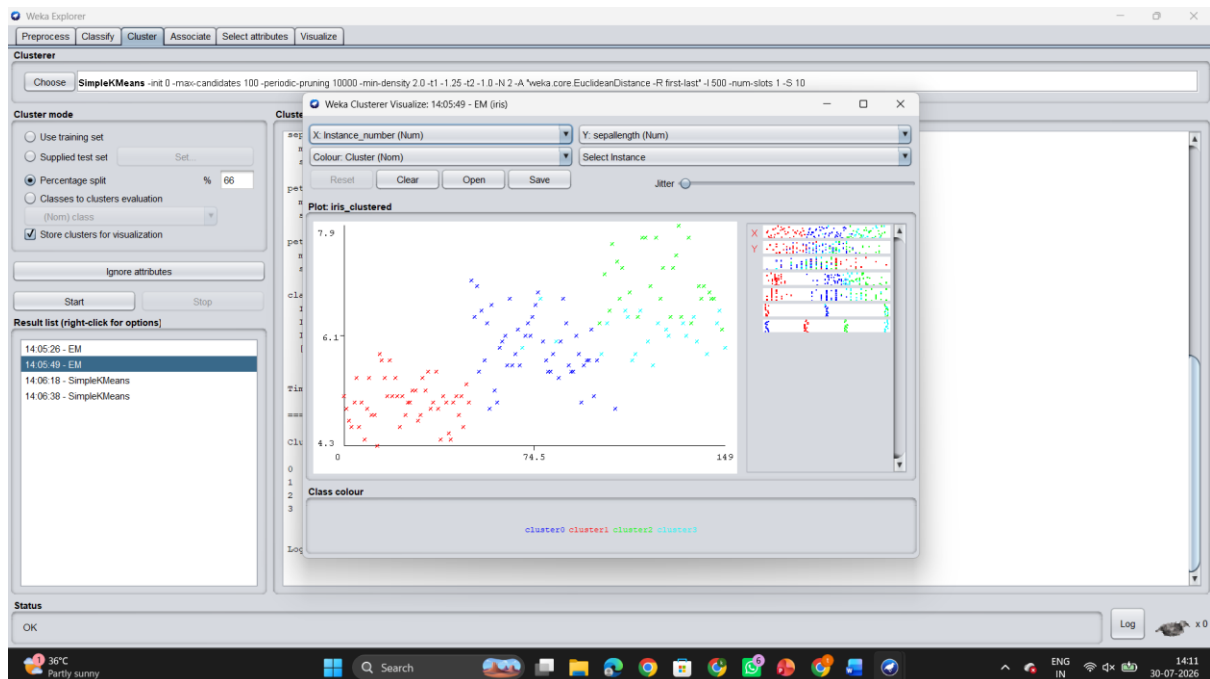
No.	Label	Count	Weight
1	Iris-setosa	50	50.0
2	Iris-versicolor	50	50.0
3	Iris-virginica	50	50.0

The 'Class' is set to 'class (Nom)' and the 'Visualize All' button is visible at the bottom right.

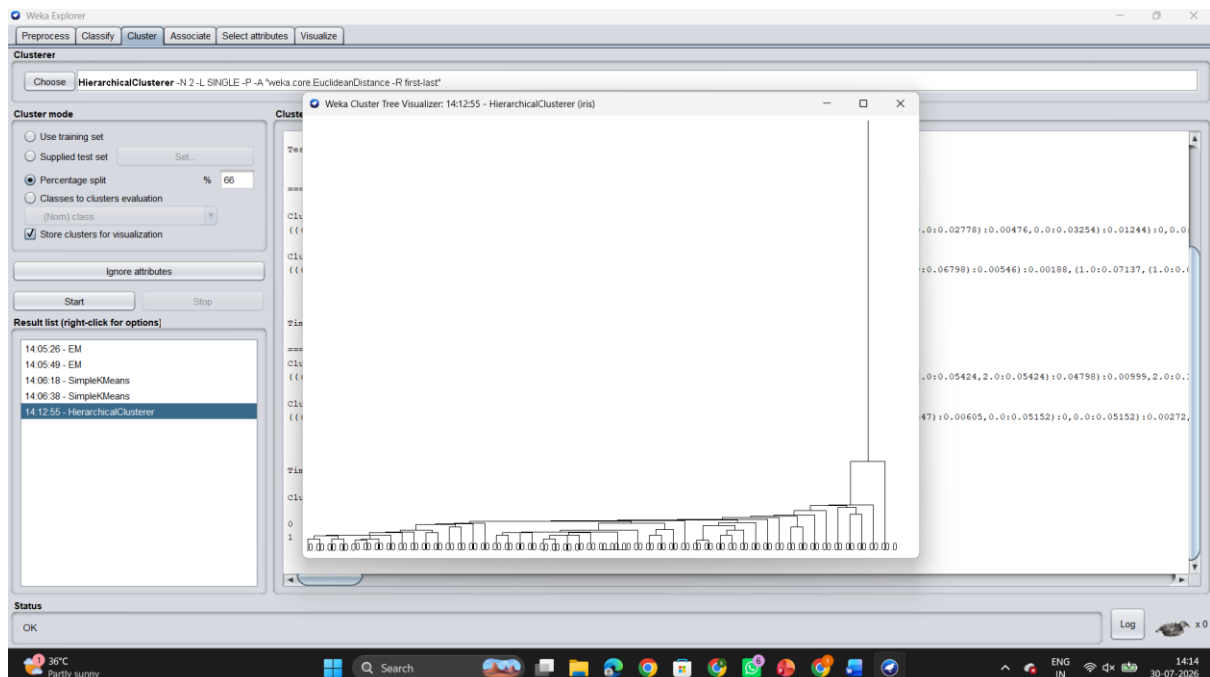
EXPERIMENT-22 DATA SEGMENTATION BY K-MEANS CLUSTER USING WEKA AND R-TOOL

The screenshot shows the Weka Explorer window with the 'Clusterer' tab active. The 'Cluster mode' is set to 'Percentage split' with a percentage of 66%. The 'Result list' on the left shows the results of the clustering process. The 'Weka Clusterer Visualize' window is open, showing a scatter plot of 'sepalength (Num)' vs 'Instance_number (Num)'. The plot displays three distinct clusters of data points, colored red, blue, and green, corresponding to the three classes of the Iris dataset. The 'Class colour' legend at the bottom indicates 'cluster0' (red), 'cluster1' (blue), and 'cluster2' (green).

EXPERIMENT- 23 DATA SEGMENTATION BY EXPECTATION MAXIMISATION ALGORITHM THROUGH WEKA



EXPERIMENT- 25 FREQUENT PATTERN MINING USING ASSOCIATION RULE THROUGH WEKA AND R-TOOLS



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S 1.0 -c 1

Start Stop

Result list (right-click):

14:21:52 - Apriori

Associator output

```
==== Associator model (full training set) ====

Apriori
=====

Minimum support: 0.15 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 12

Size of set of large itemsets L(2): 47

Size of set of large itemsets L(3): 39

Size of set of large itemsets L(4): 6

Best rules found:

1. outlook=overcast 4 ==> play=yes 4 <conf:(1)> lift:(1.56) lev:(0.1) [1] conv:(1.43)
2. temperature=cool 4 ==> humidity=normal 4 <conf:(1)> lift:(2) lev:(0.14) [2] conv:(2)
3. humidity=normal windy=FALSE 4 ==> play=yes 4 <conf:(1)> lift:(1.56) lev:(0.1) [1] conv:(1.43)
4. outlook=sunny play=no 3 ==> humidity=high 3 <conf:(1)> lift:(2) lev:(0.11) [1] conv:(1.5)
5. outlook=sunny humidity=high 3 ==> play=no 3 <conf:(1)> lift:(2.8) lev:(0.14) [1] conv:(1.93)
6. outlook=rainy play=yes 3 ==> windy=FALSE 3 <conf:(1)> lift:(1.75) lev:(0.09) [1] conv:(1.29)
7. outlook=rainy windy=FALSE 3 ==> play=yes 3 <conf:(1)> lift:(1.56) lev:(0.08) [1] conv:(1.07)
8. temperature=cool play=yes 3 ==> humidity=normal 3 <conf:(1)> lift:(2) lev:(0.11) [1] conv:(1.5)
9. outlook=sunny temperature=hot 2 ==> humidity=high 2 <conf:(1)> lift:(2) lev:(0.07) [1] conv:(1)
10. temperature=hot play=no 2 ==> outlook=sunny 2 <conf:(1)> lift:(2.8) lev:(0.09) [1] conv:(1.29)
```

Status

OK

Log

500312 +1.76%

Search

ENG IN

14:22 30-07-2026

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S 1.0 -c 1

Start Stop

Result list (right-click):

14:21:52 - Apriori

14:22:54 - Apriori

Associator output

```
=====

Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 44

Size of set of large itemsets L(2): 380

Size of set of large itemsets L(3): 910

Size of set of large itemsets L(4): 633

Size of set of large itemsets L(5): 105

Size of set of large itemsets L(6): 1

Best rules found:

1. biscuits=t frozen foods=t fruit=t total=high 788 ==> bread and cake=t 723 <conf:(0.92)> lift:(1.27) lev:(0.03) [155] conv:(3.35)
2. baking needs=t biscuits=t fruit=t total=high 760 ==> bread and cake=t 696 <conf:(0.92)> lift:(1.27) lev:(0.03) [149] conv:(3.28)
3. baking needs=t frozen foods=t fruit=t total=high 770 ==> bread and cake=t 705 <conf:(0.92)> lift:(1.27) lev:(0.03) [150] conv:(3.27)
4. biscuits=t fruit=t vegetables=t total=high 815 ==> bread and cake=t 746 <conf:(0.92)> lift:(1.27) lev:(0.03) [159] conv:(3.26)
5. party snack foods=t fruit=t total=high 854 ==> bread and cake=t 779 <conf:(0.91)> lift:(1.27) lev:(0.04) [164] conv:(3.15)
6. biscuits=t frozen foods=t vegetables=t total=high 797 ==> bread and cake=t 725 <conf:(0.91)> lift:(1.26) lev:(0.03) [151] conv:(3.06)
7. baking needs=t biscuits=t vegetables=t total=high 772 ==> bread and cake=t 701 <conf:(0.91)> lift:(1.26) lev:(0.03) [145] conv:(3.01)
8. biscuits=t fruit=t total=high 954 ==> bread and cake=t 866 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(3)
9. frozen foods=t fruit=t vegetables=t total=high 934 ==> bread and cake=t 757 <conf:(0.91)> lift:(1.26) lev:(0.03) [156] conv:(3)
10. frozen foods=t fruit=t total=high 969 ==> bread and cake=t 877 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(2.92)
```

Status

OK

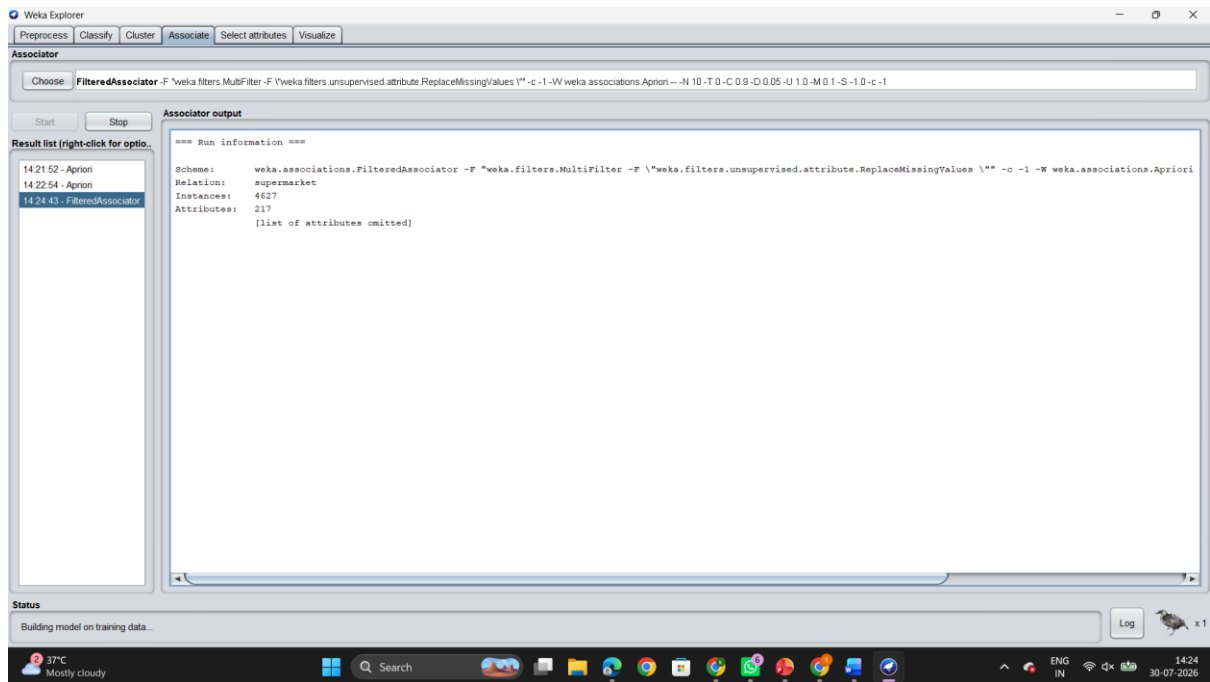
Log

500312 +1.76%

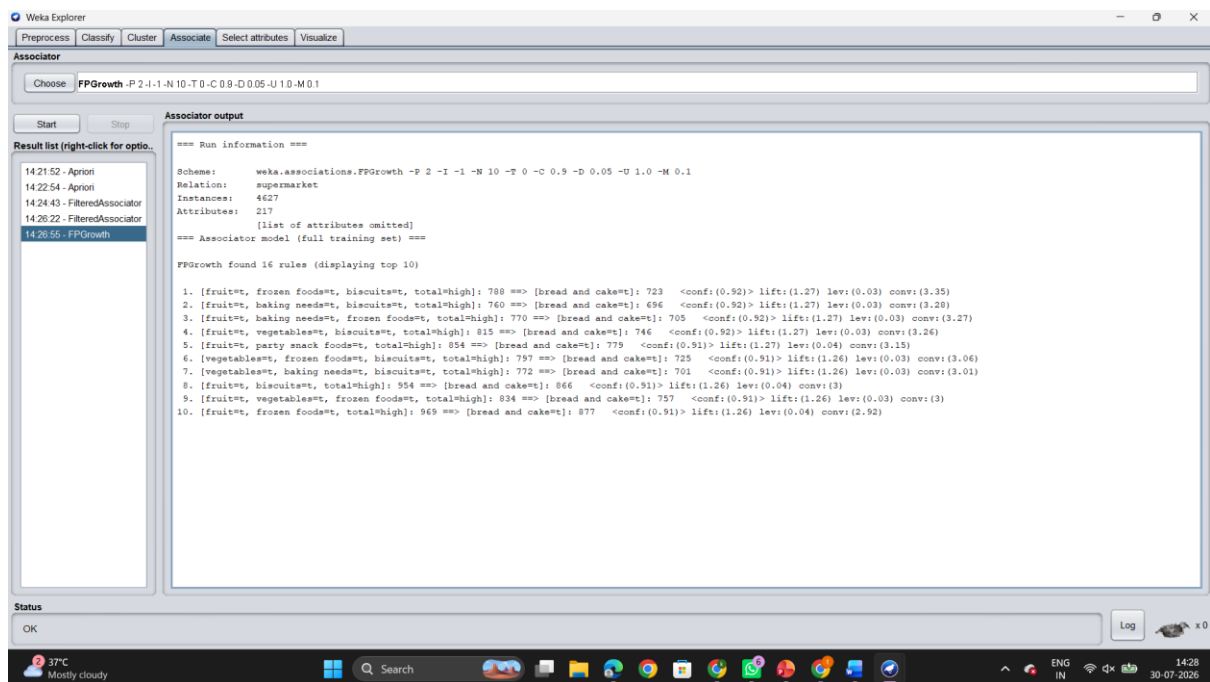
Search

ENG IN

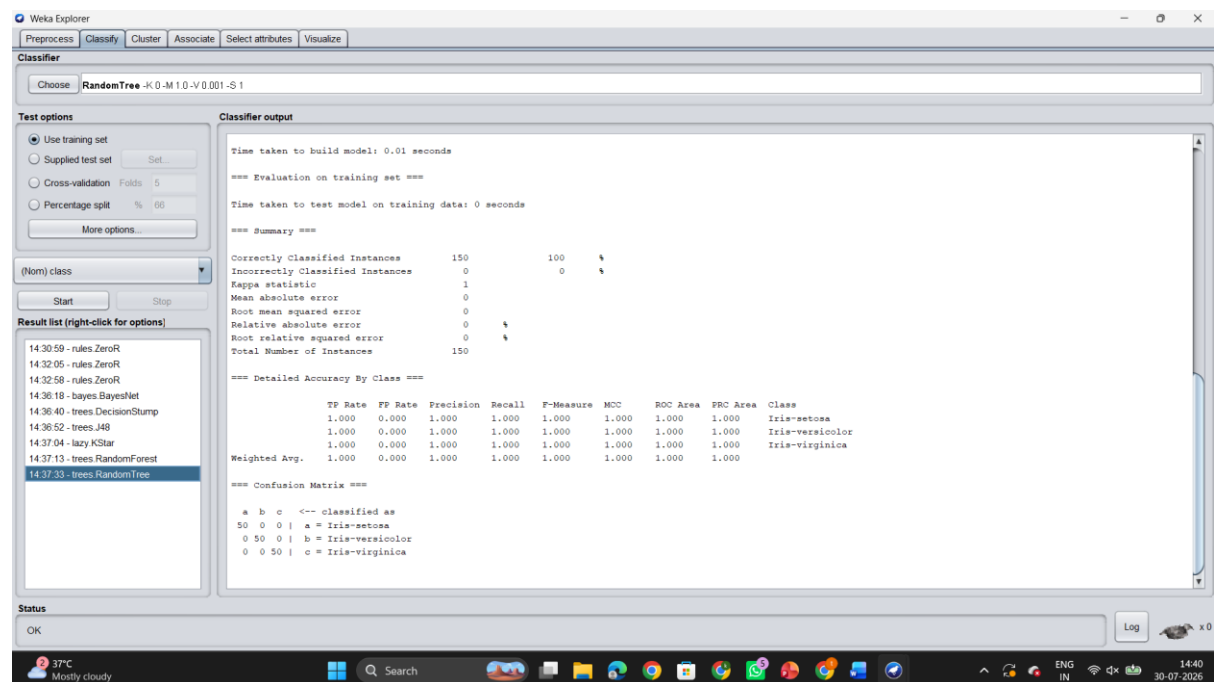
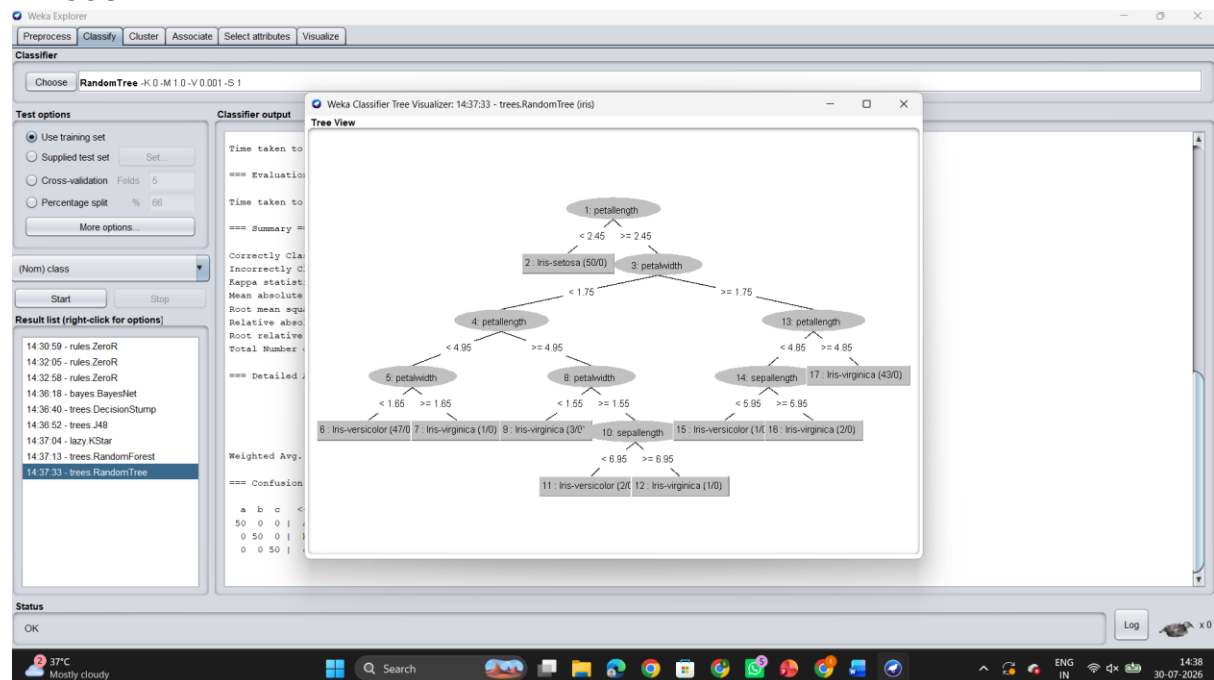
14:23 30-07-2026

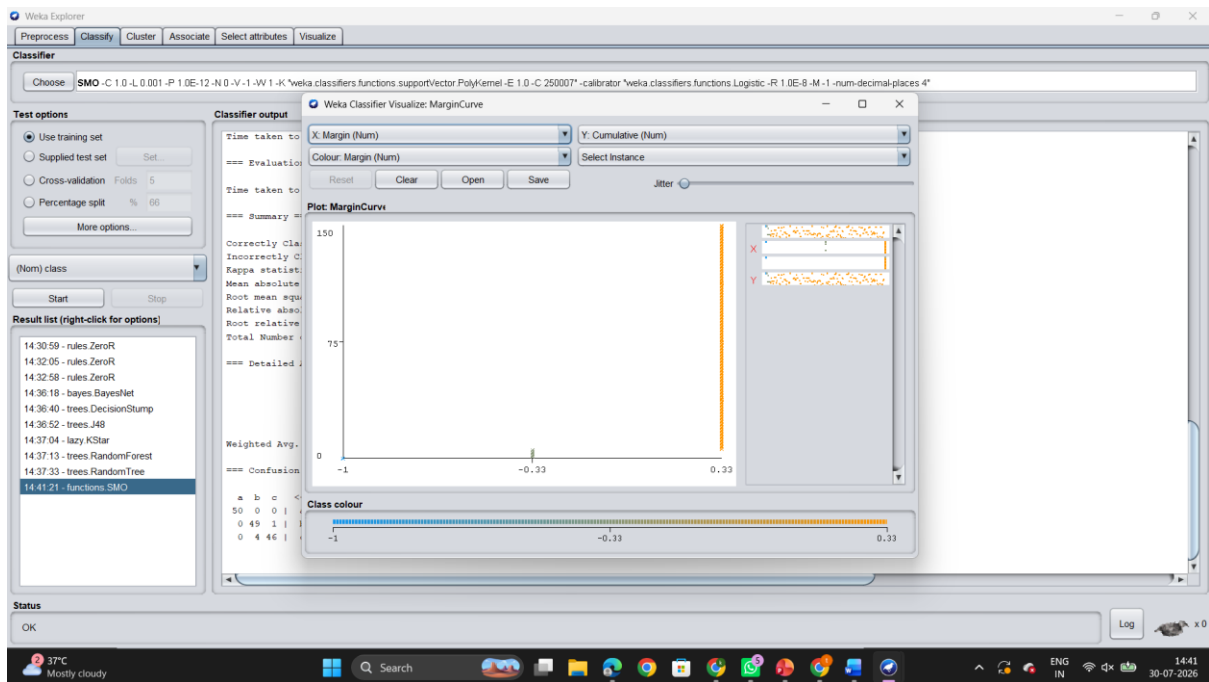


EXPERIMENT- 26 FREQUENT PATTERN MINING USING FP GROWTH THROUGH WEKA TOOL

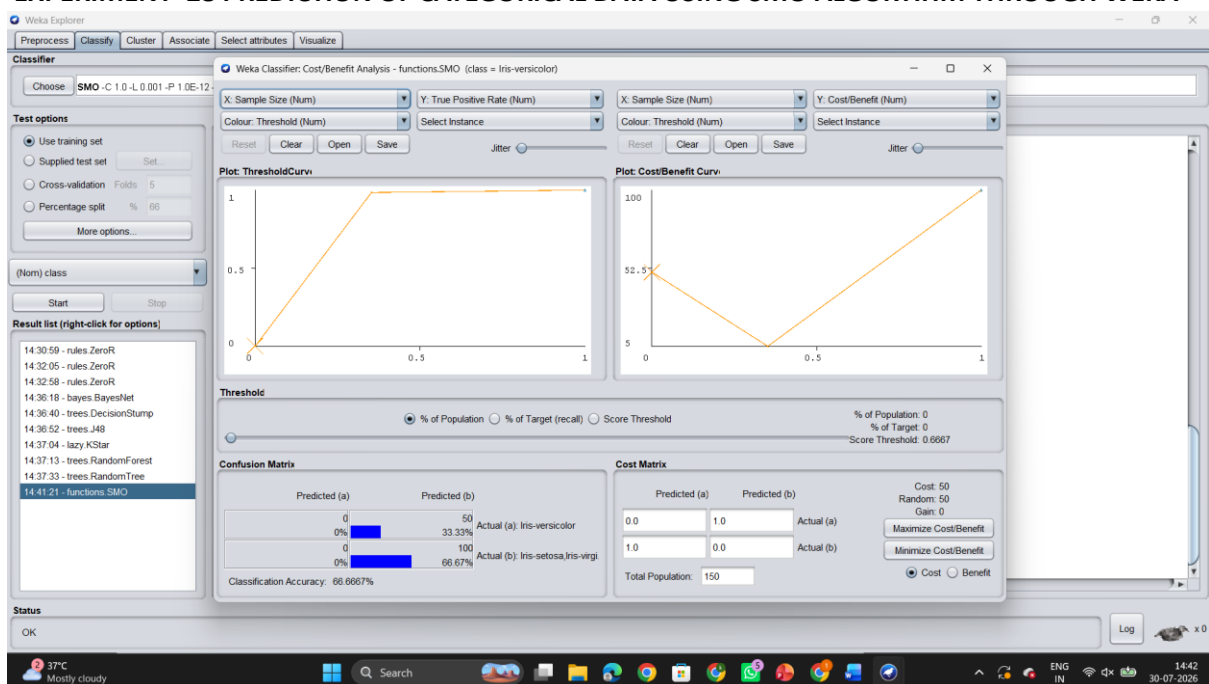


EXPERIMENT- 27 PREDICTION OF CATEGORICAL DATA USING DECISION TREE ALGORITHM THROUGH WEKA





EXPERIMENT- 28 PREDICTION OF CATEGORICAL DATA USING SMO ALGORITHM THROUGH WEKA



EXPERIMENT- 29 EVALUATING ACCURACY OF THE CLASSIFIERS

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose Logistic -R 1.0E-8 -M -1 -num-decimal-places 4

Test options

☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds 5
☐ Percentage split % 66

More options...

(Nom) class

Start Stop

Result list (right-click for options)

- 14:30:59 - rules.ZeroR
- 14:32:05 - rules.ZeroR
- 14:32:58 - rules.ZeroR
- 14:36:18 - bayes.BayesNet
- 14:36:40 - trees.DecisionStump
- 14:36:52 - trees.J48
- 14:37:04 - lazy.KStar
- 14:37:13 - trees.RandomForest
- 14:37:33 - trees.RandomTree
- 14:41:21 - functions.SMO
- 14:44:20 - functions.Logistic

Classifier output

```
petalength 0 0.0001
petalwidth 0 0

Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 143 95.3333 %
Incorrectly Classified Instances 7 4.6667 %
Kappa statistic 0.93
Mean absolute error 0.0307
Root mean squared error 0.1486
Relative absolute error 6.9153 %
Root relative squared error 31.5182 %
Total Number of Instances 150

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MDC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Iris-setosa
0.920 0.030 0.939 0.920 0.929 0.895 0.973 0.963 Iris-versicolor
0.940 0.040 0.922 0.940 0.931 0.896 0.982 0.931 Iris-virginica
Weighted Avg. 0.953 0.023 0.953 0.953 0.953 0.930 0.985 0.965

=== Confusion Matrix ===

a b c <-- classified as
50 0 0 | a = Iris-setosa
0 46 4 | b = Iris-versicolor
0 3 47 | c = Iris-virginica
```

Status

OK Log

EXPERIMENT- 30 DESCRIPTION NUMERICAL PREDICTION ANALYSIS USING LINEAR REGRESSION THROUGH WEKA

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose Logistic -R 1.0E-8 -M -1 -num-decimal-places 4

Test options

☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds 10
☐ Percentage split % 66

More options...

(Nom) class

Start Stop

Result list (right-click for options)

- 14:30:59 - rules.ZeroR
- 14:32:05 - rules.ZeroR
- 14:32:58 - rules.ZeroR
- 14:36:18 - bayes.BayesNet
- 14:36:40 - trees.DecisionStump
- 14:36:52 - trees.J48
- 14:37:04 - lazy.KStar
- 14:37:13 - trees.RandomForest
- 14:37:33 - trees.RandomTree
- 14:41:21 - functions.SMO
- 14:44:30 - functions.Logistic
- 14:47:08 - functions.Logistic
- 14:47:28 - functions.Logistic

Classifier output

```
petalength 0 0.0001
petalwidth 0 0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 144 96 %
Incorrectly Classified Instances 6 4 %
Kappa statistic 0.94
Mean absolute error 0.0287
Root mean squared error 0.1424
Relative absolute error 6.456 %
Root relative squared error 30.2139 %
Total Number of Instances 150

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MDC ROC Area PRC Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Iris-setosa
0.920 0.020 0.958 0.920 0.939 0.910 0.972 0.934 Iris-versicolor
0.960 0.040 0.923 0.960 0.941 0.911 0.972 0.934 Iris-virginica
Weighted Avg. 0.960 0.020 0.960 0.960 0.960 0.940 0.981 0.956

=== Confusion Matrix ===

a b c <-- classified as
50 0 0 | a = Iris-setosa
0 46 4 | b = Iris-versicolor
0 2 48 | c = Iris-virginica
```

Status

OK Log